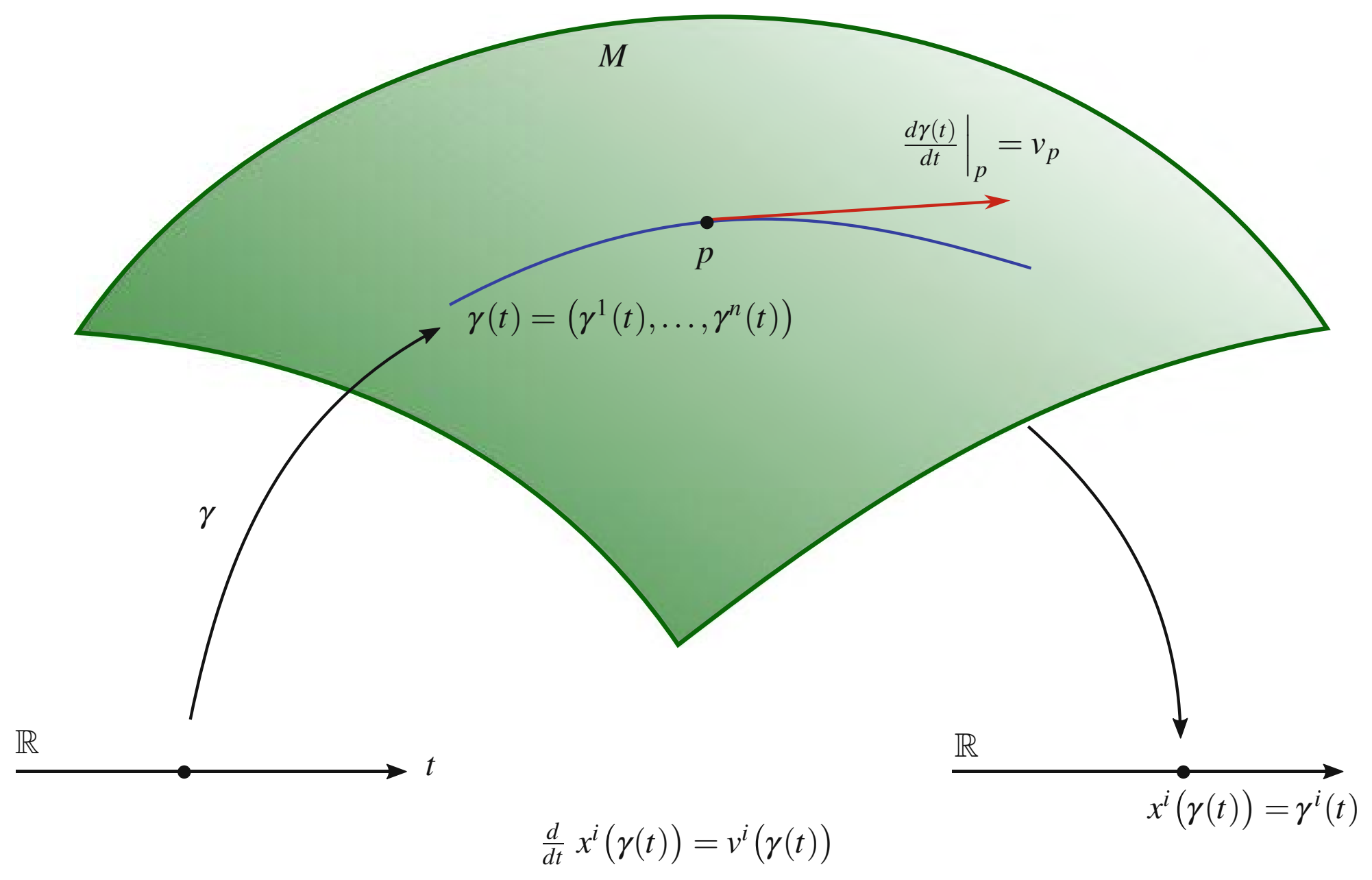

M
$\left.\frac{d\gamma(t)}{dt}\right|_p = v_p$
p
$\gamma(t) = \left(\gamma^1(t), \ldots, \gamma^n(t)\right)$
γ
$\mathbb{R}$
t
$\mathbb{R}$
$x^i\left(\gamma(t)\right) = \gamma^i(t)$
$\frac{d}{dt}\, x^i\left(\gamma(t)\right) = v^i\left(\gamma(t)\right)$